MILITARY SPECIFICATION
FOR
CASE, CARTRIDGE, 30MM, XM193

1. SCOPE

1.1 This specification covers Case, Cartridge, 30MM, XM193.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-10025	- Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-651	- Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1305

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DRAWINGS

Munitions Command

F10542940	- Case, Cartridge, 30MM, XM193
C7643674	- Classification of Cartridge Case Defects, Small Arms Ammunition.
D8889425	- Cartridge .30MM, XM552
D9204908	- Cartridge, 30MM, XM554
SKFSA9413	- Carton, Interplant Shipment, 30MM, XM193, Cartridge Cases, Empty, Unprimed
IEL10542940	- Index of Inspection Equipment Lists for: Case, Cartridge, 30MM, XM193

PUBLICATIONS

Munitions Command

AMCR 715-505	- Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge case shall comply with Drawing (Dwg) F10542940 referenced specifications, and other requirements specified herein.

3.2 First article testing.- Requirements for the submission of first article samples shall be as specified in the contract.

3.3 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Mechanical properties.- The mechanical properties specified on the applicable drawing and referenced specification shall be uniform throughout each section within specified tolerances.

3.5 Annealing.- Annealing temperature shall not exceed 1,200° Fahrenheit (F.).

3.6 Residual Stress (Mercurous nitrate) - The Cartridge Case shall be adequately stress relieved at the times and temperature suitable for development of desired properties. The cartridge case shall show no evidence of cracking after immersion in mercurous nitrate.

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3.7 Mouth Anneal.- After the final forming operation, the cartridge case shall be annealed at the mouth for a distance of 1/2 inch, minimum, 1 inch, maximum.

3.8 Neutrality.- The cartridge case surfaces shall be free of acids or alkilis.

3.9 Functional.- The cartridge case shall be subjected to a functional test when assembled into cartridge XM552, XM554 or appropriate test slug loaded to obtain average service pressure of 28.000 pounds per sq. inch, conditioned at 68° to 72° Fahrenheit (F). The average velocity of the cartridge conditioned at 68° to 72° Fahrenheit (F) shall be 2200 feet per second (Ft/sec) plus or minus 20 ft/sec at 78 feet from the muzzle of the test barrel.

3.9.1 Soundness.- After firing, the cartridge case shall be free of splits or ruptures.

3.9.2 Extraction.- After firing, the cartridge case shall be extracted by the service extractors when the weapon is operated in the normal manner.

3.9.3 Primer retention.- The cartridge case shall retain the primer during firing and extraction.

3.10 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects.- The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches and porosity.

3.10.1.1 Flow of metal.- The flow of metal in the section of the cartridge case joining the head and sidewall shall follow the direction of the contour of the interior curved surface without presence of folds, cold shuts or roughness.

3.10.2 Burr.- No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile, or which might be injurious to personnel handling the cartridge case.

3.10.3 Foreign matter.- No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed

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4.1 (Cont'd)

necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

a. Inspection and non-firing test(s)

100 cartridge cases

b. Firing test(s)

200 cartridge cases

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.2 Lot identification.- Each cartridge case and each packed cartridge case lot shall be identified in accordance with applicable drawings, supplemented as directed by the procuring agency.

4.3.2 Examination.- Examination for major and minor defects shall be performed as specified herein. Each cartridge case, found to be defective, shall be rejected.

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4.3.2.1 Sampling and acceptance criteria.- Sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235, except as otherwise indicated. Defect characteristics may be grouped into a class and considered collectively, or defect characteristics may be considered individually to determine acceptability. When acceptability is determined on a class basis, in addition to the AQL assigned to the major class, an AQL shall be assigned to individual characteristics within the major class, limiting the number of defectives containing a particular major defect to less than the number of defectives permitted for the class. However, when three or less defect characteristics are listed in a classification or when sampling procedures of MIL-STD-1235 are used, acceptability shall not be determined on a class basis for major defects.

4.3.2.2 AQL's.- The following AQL's are assigned:

a. Individual defect characteristic basis.

Major defects	0.25%
Minor defects	0.40%

b. Defect characteristic, class basis.

Major defects	1.50%
Minor defects	2.50%

c. Individual defect characteristic associated with the class basis.

Major defects	0.40%

4.3.3 Classification of defects 1/. The classification of defects shall be as follows:

4.3.3.1 Case, Cartridge (see Dwg. F10542940).

Categories and defects	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Corrosion	Visual
103. Oil or grease	Visual
104. Head configuration (round head or beveled underside)	Visual

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Categories and defects		Method of Inspection
MAJOR	(Cont'd)	
105.	Chamfer, head rim, missing	Visual
106.	Profile, maximum	Gage
107.	Total length, maximum	Gage
108.	Diameter formed cylindrical section, min.	Gage
109.	Length from base to minimum diameter of formed cylindrical section, maximum	Gage
110.	Diameter, middle body taper, minimum	Gage
111.	Length to basic diameter on belt	Gage
112.	Thickness, head rim	Gage
113.	Diameter, rim	Gage
114.	Concentricity, rim diameter with body	Gage
115.	Diameter, extractor groove, maximum	Gage
116.	Concentricity, extractor groove diameter with body	Gage
117.	Width, extractor groove minimum	Gage
118.	Diameter, primer pocket	Gage
119.	Concentricity, primer pocket with body	Gage
120.	Depth, primer pocket	Gage
121.	Diameter, flash tube hole	Gage
122.	Concentricity flash tube hole with primer pocket	Gage
123.	Thickness, wall at .500 inches from inside base	Gage
124.	Diameter, inside mouth, minimum for minimum length	Gage
125.	Diameter, belt	Gage
126.	Concentricity of head	Gage
MINOR		
201.	Foreign matter, other than oil, grease or corrosion	Visual
202.	Scratch (other than die scratch)	Visual
203.	Fold, wrinkle or buckle	Visual
204.	Radii or chamfers missing or incorrect	Visual
205.	Total length, minimum	Gage
206.	Thickness, wall, at mouth	Gage
207.	Diameter, extractor groove minimum	Gage
208.	Width, extractor groove maximum	Gage
209.	Thickness, base	Gage
210.	Diameter, inside mouth, maximum	Gage
211.	Depth, indented head (alternative design)	Gage
212.	Marking, missing, incorrect or illegible	Visual
213.	Surface finish	Visual

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MIL-STD-651 shall apply in defining and evaluating visual defects. No photographs of 30mm cases are included. However, the defect classification of the 20mm M103 Brass Case will serve as a guide for inspection.

4.3.4 Tests.- The tests listed in table I shall be conducted on each sample in accordance with the applicable methods and procedures prescribed in 4.4. For acceptance, the results of each test shall comply with the applicable requirement(s).

Table I

Test		Sample Size	Requirement Paragraph
Hardness	1/	5	3.4
Residual stress	2/	10	3.7
Neutrality	3/	10	3.8
Functional	4/	200	3.9
Flow of metal	5/	5	3.10.1.1

1/ Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be subjected to this test. Failure of two or more of the sample cartridge cases to comply with the hardness requirements shall cause rejection of the lot or sample. If one of the samples fails to comply, a second sample of 10 cartridge cases shall be randomly selected from the lot and tested. The lot or sample shall be rejected if a total of two or more cartridge cases of the combined first and second samples fail to comply with the applicable hardness requirement. Failure will be considered failure of the average of the multiple hardness determinations made at each position to meet the requirement for that position.

2/ Ten cartridge cases from the initial production sample and ten cartridge cases from each production lot shall be subjected to this test. The lot or sample shall be rejected if one or more cartridge cases of the lot or sample split.

3/ Ten cartridge cases from the initial production sample and ten cartridge cases from each production lot shall be subjected to this test. The lot or sample shall be rejected if one or more cartridge cases show evidence of acidity or alkalinity.

4/ Two hundred cases from the initial production sample and two hundred cartridge cases from each production lot shall be subjected to this test. The lot or sample shall be rejected if any malfunction or casualty of table II occurs in number(s) equal to or greater than the applicable "Rej" number. If malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 400 cartridge cases randomly selected from the lot or sample shall be tested. The lot or sample shall be rejected if in the combined samples malfunctions or casualties occur in number(s) equal to or greater than the applicable "Rej" number.

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5/ Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be subject to this test. Failure of one or more cartridge cases to comply with the requirements of 3.10.1.1 shall be cause of rejection of the initial production sample or the production lot. The cartridge cases used for the hardness test may be used for this test.

Table II

Firing Casualties	Acc	Rej
Failure to extract 1/	0	1
Blown primer 2/	0	1
Splits and ruptures 3/		
Splits:		
I	6	17
J	2	12
K, L, or M	0	1
Rupture, partial		
J, or K	1	6
L	0	1
Circumferential rupture, complete	0	1
Detached metal 4/	0	1

1/ A fired sample cartridge case remaining in the chamber after the normal extraction cycle will be classed defective, unless it can be shown that the malfunction was caused by a defective weapon. Usually, "failure to extract" is associated with "soft" metal, resulting in non-resilient obturation and rim shearing of the cartridge case.

2/ The sample cartridge case shall be classed defective only if it is evident by visual inspection that both the primer pocket and case head are enlarged and distorted.

3/ Refer to Dwg. C7643674 for classifying splits and ruptures resulting from the function and casualty test. If a split or partial rupture extends into two or more defined areas, only the most severe defect criterion of Table II for the areas involved shall be used. A sample cartridge case which separates into two or more portions, shall be classed defective as a complete circumferential rupture.

4/ The case lot will not be penalized if malfunction cannot be shown to have resulted from a defective case. There must be evidence of detached metal from fired case exterior. The lot will not be penalized for the detachment of metal shavings from the crimped area of the case interior.

4.3.4.1 Unlisted firing defects.- The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification, occurring in any firing test, indicates that the

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product is unsuited for the purpose intended.

4.3.4.2 Process control.- Monitoring at unscheduled intervals shall be conducted for the following process operations to determine compliance with the following requirements.

Anneal	3.5
Stress relief	3.6
Mouth Anneal	3.7

Failure to comply with requirements shall be cause for rejection of all units processed since the last monitored approval of the process.

4.3.5 Packaging and marking inspection.- Inspection for packing and marking shall be in accordance with MIL-P-10025.

4.3.6 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-10542940.

4.4 Test methods and procedures.

4.4.1 Hardness.- The cases of the test sample(s) will be sectioned as shown on Dwg. F10542940. The head sections will have approximately parallel surfaces finished to 63 microinches or finer, care being taken not to change the hardness in obtaining this finish. Four determinations each will be made at the indicated body position for each sample case, two on each half on the interior as shown, and an average being computed for each position. Two determinations each will be made at the indicated head position as shown for each sample case and an average computed for each position. The method of test will be as prescribed in Federal Test Method Standard 151.

4.4.2 Residual stress.-(Mercurous nitrate).- The method of test shall be as prescribed in AMCR 715-505 Vol. 8 for case mercurous nitrate test.

4.4.3 Neutrality.- The method of testing cartridge cases for acidity and alkalinity will be as follows using red and blue sensitive litmus paper. Neutral distilled water will be used to moisten the paper and the case to provide adhesion. The litmus paper will be handled with forceps to prevent contamination from fingers. Smoking should not be permitted in the vicinity of test since tobacco smoke will turn red litmus blue. Pieces of red and blue sensitive litmus paper will be placed in several locations both on the interior and exterior of the case. A definite color change is defined as the color produced when blue litmus is immersed for thirty seconds in one thousandth normal (.001N) hydrochloric or sulfuric acid, or when red litmus is immersed for thirty seconds in one thousandth normal (.001N) sodium hydroxide. Litmus paper is considered of sufficient sensitivity if it shows change in color when immersed for 45 seconds, with stirring, in five ten thousandths normal (.005N) solutions of the corresponding acid or alkali mentioned.

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4.4.4 Functional.- A propellant charge will be established which will meet the velocity and chamber pressure requirements of 3.9. Velocity and pressure test equipment listed on IEL-10542940. The two hundred (200) test sample cases will be assembled into cartridges, 30mm XM552 or XM554, and fired in a XM140 Machine Gun in bursts of 50 rounds at a cyclic rate of approximately 450 rounds per minute. The method of test shall be as prescribed in AMCR 715-505 Vol. 8. The gun barrel used will be at ambient temperature at the beginning of the test and will be cooled to ambient temperature between bursts. This test will be conducted at a facility designated by the procuring activity.

4.4.5 Flow of metal.- The cartridge case shall be sectionalized through the center of the head, polished and examined for folds and cold shuts in the radius area of the head, in accordance with the procedures for this test prescribed by Federal Test Method Standard No. 151. One hundred (100X) minimum magnification shall be used for this examination.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level C.- Cartridge cases shall be packaged in accordance with Dwg. SK FSA 9413.

5.2 Marking and labeling.- Marking on cartons shall be in accordance with MIL-P-10025.

5.3 Data cards.- Data cards shall comply with the requirements of MIL-STD-1167. When data cards are specified by the contract, they shall be prepared and distributed by the contractor.

6. NOTES

6.1 Ordering data.- Invitation for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 First article sample requirements (see 3.2 and 4.2).

6.1.4 Packing instructions (see 5.1).

6.1.5 Provision for the submission of a manufacturing procedure (see 4.2).

6.1.6 Data card requirements (see 5.3).

6.1.7 Instructions for shipment of first article inspection samples for function and casualty test.

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6.1.8 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance test purposes.

6.1.9 Provision for the submission of contractor inspection reports containing final inspection results for each lot of metal parts presented to the Government.

6.1.10 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

6.2 AQL's.- The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.3 Intermediate point inspection.- The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality in the end item.

6.4 First article sample firing test.- The contracting officer will notify the testing facility when to proceed with firing tests.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project no. 1305-A470

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